

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Third Semester of B.Tech. Theory Examination(EC)
March-April 2018

MA243 Transforms and Matrix Algebra

Date: 26/03/2018 (Monday)

Time: 01:30 pm to 04:30 pm

Maximum Marks: 70

Instructions:

- i. The question paper contains two sections.
- ii. Section I and II must be attempted in separate answer sheets.
- iii. Figures to the right indicate marks.
- iv. Make suitable assumptions and draw neat figure where it is required.
- v. All notations and terminologies are standard.

Section-I

Q-1 Choose the correct answer from the given options in the following: [07]

(1) The period of $\sin x$ is _____. 01
 (a) π (b) $-\pi$ (c) 2π (d) -2π

(2) The value of an integral $\int_0^{2\pi} \cos x \, dx$ is _____. 01
 (a) 0 (b) 1 (c) -1 (d) 2π

(3) Which one of the following is an even function of t ? 01
 (a) $t^2 - 3t$. (b) t^2 . (c) $\sin(5t)$ (d) $t^3 - 5$.

(4) The value of $L[f(t-a)u(t-a)]$ is _____. 01
 (a) $e^{as}F(s)$ (b) $e^{-as}F(s)$ (c) $e^{at}F(t)$ (d) $e^{-at}F(t)$

(5) The value of $L[\sinh(2t)]$ is _____. 01
 (a) $\frac{2}{s^2 - 4}$ (b) $\frac{2}{s^2 + 4}$ (c) $\frac{2}{s^2 - 2}$ (d) $\frac{2}{s^2 + 2}$

(6) The Laplace transform of a function f exists if the function 01
 (a) f is piecewise continuous. (b) f is of exponential order.
 (c) f is piecewise continuous and of exponential order. (d) f is neither piecewise continuous nor of exponential order.

(7) The inverse Fourier transform of a function f is 01

(a) $f(x) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} F(s) e^{isx} ds.$ (b) $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{isx} ds.$

(c) $f(x) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} F(s) e^{-isx} ds.$ (d) $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds.$

Q-2 Attempt any Four. [16]

- (1) Find the Fourier series of $f(x) = \frac{\pi - x}{2}$ in the interval $[0, 2\pi]$. 04
 - (2) Expand $f(x)$ in a Fourier series, where $f(x) = \begin{cases} 0 & ; -2 < x < 0 \\ 1 & ; 0 < x < 2 \end{cases}$. 04
 - (3) Obtain the Fourier series of $f(x) = x^2$, where $-1 < x < 1$. 04
 - (4) Using Convolution theorem find the inverse Laplace transform of $\frac{1}{s^2(s-1)}$. 04
- Find the Laplace transform of the function $f(t)$ defined by 04
- (5) $f(t) = \begin{cases} 3 & ; 0 < x < 2 \\ 0 & ; 2 < x < 4 \end{cases}$, $f(t+4) = f(t)$. 04
 - (6) Evaluate $\int_0^{\infty} \frac{x^2}{(x^2+1)^2} dx$ using Parseval's identity for the Fourier transform. 04

Q-3 (a) Attempt any Two. [06]

- (1) Define the half range Sine and half range Cosine Fourier series. 03
- (2) Solve the differential equation $\frac{d^2y}{dt^2} + y = 1$ with $y(0) = 1$ and $y'(0) = 0$ using Laplace transform. 03
- (3) Find the Laplace transform of $t^3 + 3t^2 + \sin(2t) + e^t \cos(2t)$. 03

Q-3 (b) Define the Fourier transform and find it for the function [06]

$$f(x) = \begin{cases} 1 - x^2 & ; |x| \leq 1 \\ 0 & ; |x| > 1 \end{cases}$$

OR

Q-3 (b) Define the Fourier sine and cosine transform. Find the Fourier sine and cosine transform of $f(t) = e^{-at}$. [06]

Section-II

Q-4 Choose the correct answer from the given options in the following: [07]

- (1) If $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of a matrix $\begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$, then the corresponding eigenvalue is _____. 01
- (a) 5 (b) 2 (c) -2 (d) Cannot be determined

- (2) If 0 is an Eigen value of a matrix A , the value of $\det(A)$ is _____. **01**
 (a) 1 (b) 0 (c) 2 (d) -1
- (3) The Z-transform $U(z)$ of a discrete time signal $u(n)$ is defined as **01**
 (a) $\sum_{n=-\infty}^{\infty} u(n)z^n$ (b) $\sum_{n=-\infty}^{\infty} u(n)z^{-n}$ (c) $\sum_{n=0}^{\infty} u(n)z^n$ (d) None of the mentioned
- (4) If $U(z)$ is the Z-transform of signal $u(n)$, then the Z-transform of $\delta(n)$ is _____. **01**
 (a) 1 (b) 0 (c) -1 (d) Not possible
- (5) According to the shifting property of Z-transform if $U(z)$ is the Z-transform of signal $u(n)$, then what is the Z-transform of u_{n-k} ? **01**
 (a) $z^k U(z)$ (b) $z^{-k} U(z)$ (c) $U(z-k)$ (d) $U(z+k)$
- (6) If $r = x\hat{i} + y\hat{j} + z\hat{k}$, then $\text{div } r$ equal to _____. **01**
 (a) 3 (b) r (c) 0 (d) $-r$
- (7) If a vector $a = \frac{i}{\sqrt{3}} + \frac{j}{\sqrt{3}} + \frac{k}{\sqrt{3}}$, then modulus of vector a is _____. **01**
 (a) 1 (b) 0 (c) -1 (d) 2

Q-5 Attempt any Four.

[16]

- (1) Find the eigenvalues and eigenvectors for the matrix $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$. **04**
- (2) Using Cayley -Hamilton theorem reduce $2A^5 - 3A^4 + A^2 - 4I$ into a linear polynomial in A , where $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$. **04**
- (3) Find Z-transform of $\sin n\theta$ and $\cos n\theta$. **04**
- (4) Find the Z-transform of (i) a^n (ii) $(n+1)^2$. **04**
- (5) Find the inverse Z-transform of $\frac{4z}{z-a}$ for (i) $|z| > |a|$ (ii) $|z| < |a|$. **04**
- (6) If $\vec{r} = (a \cos \eta t)\hat{i} + (b \sin \eta t)\hat{j}$, where a, b and η are constants then prove that **04**

$$\frac{d^2 \vec{r}}{dt^2} + \eta^2 \vec{r} = 0.$$

Q-6 (a) Attempt any Two.

[06]

(1) Evaluate Curl of the vector $\vec{F} = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$ at (1, -1, 1).

03

(2) A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$, where t is the time.

03

Find the component of its velocity and acceleration at time t=1.

(3) Find the directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at point (2,1,3) in the direction of the vector $a = i - 2k$.

03

Q-6

(b) Find the characteristic equation of a matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence obtain

[06]

$$A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I.$$

OR

Q-6

(b) Find the eigenvalues and eigenvectors for the matrix $A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$.

[06]
